Auto Ticket Classification using Flow Designer

ServiceNow Project Report

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Phase 1: Brainstorming & Ideation Phase

Problem Statement

In a school IT help desk, tickets arrive with different issues such as network problems, hardware faults, software installation or account access. Classifying and routing each ticket manually is slow, inconsistent and delays resolution.

Proposed Idea

Use ServiceNow Flow Designer to automatically classify every newly created School IT ticket and assign it to the right group, with no manual triage.

Ideas Considered

Idea	Pros	Cons
Manual classification by staff	Simple, no setup	Slow, error-prone
Business Rules (scripting)	Flexible	Needs code, harder to maintain
Flow Designer (selected)	Low-code, visual, easy to test and maintain	Limited to available actions

Expected Outcomes

- Faster ticket handling
- Consistent categorisation
- Tickets routed to the correct assignment group automatically

Phase 2: Requirement Analysis Phase

Functional Requirements

- A custom table to store School IT tickets.
- Auto-numbering for every ticket.
- A flow that triggers when a ticket is created.
- The flow sets Category, Subcategory and Assigned Group based on the ticket's short description / description.
- All work captured in a Local Update Set.

Non-Functional Requirements

- Easy to maintain (low-code, no complex scripts).
- Runs instantly on ticket creation.
- Reliable and repeatable.

Table Fields Required

Field	Purpose
Number	Auto-generated unique ticket ID
Caller	Person who raised the ticket
Category	Main type of issue (set by the flow)
Subcategory	Detailed type of issue
Short Description	One-line summary
Description	Full issue details
State	Current status of the ticket
Assigned Group	Team handling the ticket
Assigned to	Individual technician

Tools & Environment

- ServiceNow Personal Developer Instance (dev185977)
- Flow Designer
- Local Update Set: Project Update Set

Phase 3: Project Design Phase

Solution Architecture

User creates ticket → Record created in Incident WorkFlow table → Flow trigger fires → Flow checks ticket details → Category, Subcategory and Assigned Group are updated → Ticket is ready for the technician.

Data Design

Property	Value
Table label	Incident WorkFlow
Table name	u_incident_workflow
Auto Number	Enabled
Fields	Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, Assigned to

Flow Design

Element	Design
Flow name	Auto Classify School IT Tickets
Trigger	Record → Created (on Incident WorkFlow table)
Logic	Conditions on Short Description / Description decide the category
Actions	Update Record to set Category, Subcategory and Assigned Group

Classification Logic (Sample)

If ticket mentions...	Category	Assigned Group
wifi, internet, network	Network	Network Team
laptop, printer, projector	Hardware	Hardware Support
install, software, application	Software	Software Support
password, login, account	Account Access	Service Desk

Phase 4: Project Planning Phase

The project is divided into small milestones and completed step by step.

Step	Milestone	Output
1	Create Local Update Set	Project Update Set (Global)
2	Create custom table and enable Auto Number	Incident WorkFlow table
3	Create required fields	9 fields on the table
4	Build flow with Created trigger	Auto Classify School IT Tickets flow
5	Add classification conditions and update actions	Working flow logic
6	Test with sample tickets	Test results
7	Prepare documentation and upload to GitHub	Public repository
8	Record demo video and upload to Google Drive	Public Drive link

Risks & Mitigation

Risk	Mitigation
Flow not triggering	Verify trigger table and that the flow is activated
Wrong classification	Refine conditions and re-test with more samples
Changes not captured	Keep Project Update Set as the current update set throughout

Phase 5: Project Development Phase

Step 1: Create the Update Set

- Go to System Update Sets → Local Update Sets → New.
- Name: **Project Update Set**, Application: **Global**.
- Set it as the current update set.

Step 2: Create the Table

- Go to System Definition → Tables → New.
- Label: **Incident WorkFlow** (u_incident_workflow).
- Enable **Auto Number** so each ticket gets a unique Number.

Step 3: Create the Fields

- Add: Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, Assigned to.

Step 4: Build the Flow

- Open Flow Designer → New → Flow.
- Name: **Auto Classify School IT Tickets**.
- Trigger: **Created** on the Incident WorkFlow table.
- Add conditions (If / Else If) on Short Description and Description.
- In each branch add an **Update Record** action to set Category, Subcategory and Assigned Group.
- Save and **Activate** the flow.

Phase 6: Project Testing Phase

Test cases are executed by creating new tickets and checking the flow execution in Flow Designer → Executions.

#	Test Input (Short Description)	Expected Result	Status
1	Wifi not working in lab	Category: Network, Group: Network Team	Pass / Fail
2	Printer not printing	Category: Hardware, Group: Hardware Support	Pass / Fail
3	Need software installed	Category: Software, Group: Software Support	Pass / Fail
4	Forgot my password	Category: Account Access, Group: Service Desk	Pass / Fail
5	Unrelated / unclear text	Handled by default branch	Pass / Fail

What to Verify

- Auto-number is generated on ticket creation.
- Flow execution shows status Completed.
- Category, Subcategory and Assigned Group are filled correctly.
- No errors in the flow execution log.

Fill the Status column after testing and attach screenshots in the repository.

Phase 7: Project Documentation Phase

All project files and documents are uploaded to a **Public GitHub Repository**, organised phase-wise.

Suggested Repository Structure

```
Auto-Ticket-Classification/
  README.md
  1_Brainstorming_and_Ideation_Phase/
  2_Requirement_Analysis_Phase/
  3_Project_Design_Phase/
  4_Project_Planning_Phase/
  5_Project_Development_Phase/
  6_Project_Testing_Phase/
  7_Project_Documentation_Phase/
  8_Project_Demonstration_Phase/
```

Documents to Include

- Project report (this document)
- Screenshots of update set, table, fields, flow and test results
- Exported Update Set XML (optional)
- README with project name, purpose and links

Repository Checklist

- Repository visibility is **Public**.
- Every phase folder contains its files.
- README explains the project clearly.

Phase 8: Project Demonstration Phase

A complete demo video is recorded and uploaded to **Google Drive** with sharing set to **Anyone with the link can view**.

Demo Video Must Include

- Screen sharing of the project
- Student's voice-over / explanation
- Project name
- Purpose of the project

Suggested Demo Flow

Step	What to show
1	Introduce yourself, the project name and its purpose
2	Show the Project Update Set
3	Show the Incident WorkFlow table and its fields
4	Open the flow in Flow Designer and explain trigger and conditions
5	Create a new ticket live
6	Show the flow execution and the auto-classified ticket
7	Conclude with benefits and outcome

Submission Links

Item	Link
GitHub Repository (Public)	_____
Demo Video (Google Drive)	_____

Conclusion: The project automates ticket classification and routing in ServiceNow using Flow Designer, reducing manual effort and improving response time.